

Reading a neuron’s tuning from its correlations with other neurons

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Abstract

What a neuron represents is usually established with a stimulus. We ask whether the same facts are contained in the correlation matrix of a recorded population, which keeps the pairwise relations among the neurons and discards the stimulus. A transformer sees the correlation matrix of 512 sampled neurons and nothing else: no labels, and no neuron it was trained on. On in-vivo MICrONS responses it decodes preferred orientation to a mean error of 25° , against 45° for an uninformed decoder and 41° for the best constant guess. Averaged over eight orientation bins rather than over neurons the error is 28° , and with training and test matrices built from disjoint halves of the movie it is 34° . Asked which cardinal axis a neuron is closer to, it is right for 79% of neurons, against 56% for the majority answer; the obliques are not resolved. It decodes receptive-field position, as fitted by the digital twin, at $R^2 = 0.23$. On the digital-twin substrate the three figures are 20° (24.5° balanced), 87% and 0.48. The relations between orientation classes are close to circulant, which fixes the classes up to rotation and reflection; only a departure from that symmetry fixes the rotation of the frame. The departure survives equal numbers of neurons per orientation, and relations built from disjoint halves of the movie, which share its orientation content. The candidate departure is an anisotropy of the signal correlations: in V1, the neurons preferring 90° have the most similar responses to one another. A model population with that anisotropy alone and equal numbers per orientation recovers the frame; the anisotropy could not be removed from the cortical data, and a cardinal bias in the movie’s orientation content, which also suffices in the model, was not measured. Across 33 mice of the Allen Brain Observatory, with grating relations, orientation transfers to unseen animals only from populations that mix animals. An Allen animal supplies 171 to 633 cells, so its populations of 128 neurons overlap in most of their neurons, and they are not decoded measurably below the best constant. For receptive fields, what crosses animals is each animal’s mean receptive-field position, and only weakly the position of a neuron relative to its mouse’s mean.

1 Introduction

A neuron in visual cortex can be characterised by what it responds to: vertical gratings, or the upper left of the screen. This is conventionally established with the stimulus. We ask whether the same facts are contained in a description that mentions no stimulus. For a population that watched the same movie, one can compute the correlation of every pair of neurons and then discard the movie. Does the resulting matrix determine which neuron prefers which orientation, and where each neuron’s receptive field lies?

The idea that content can be read from relations is old. Shepard and Chipman proposed that a representation need not resemble what it stands for as long as the similarities among representations mirror the similarities among the things represented [1], and representational similarity analysis turned this into a method that compares brains and models through their similarity structures rather than their coordinates [2]. Structuralist accounts of experience claim that what a state represents, or how it feels, is fixed by its position in a space of relations [3].

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Lässig [4] formulates the corresponding question for neural networks and gives the decoding framework we use here.

Two lines of work come close. Tsodyks et al. [5] showed that the spike-triggered average of spontaneous population activity reproduces the map of a neuron’s preferred orientation, with no stimulus present; the tuning was read from that pattern by comparison with the evoked map, a stimulus-labelled reference. Unsupervised manifold methods recover a neuron’s tuning from the structure of population activity alone, without measuring the variable it encodes [6], but the recovered coordinate is defined only up to the symmetries of the latent space, and it is related to the measured variable by aligning to it afterwards. In both cases the relations locate a neuron within the structure of the encoded variable, and something outside the relations, the evoked map or the measured behaviour, says which value that location corresponds to. We ask whether the relations among neurons fix the content of each neuron when the stimulus is discarded and the population being read contains no labelled neuron, no map and no behaviour to compare with: our decoder is trained on labelled neurons, but they are other neurons, and on Allen other animals. We also ask whether that content is readable across thousands of individually identified neurons and across animals. The nearest machine-learning work reads properties of a neuron from population activity: NuCLR learns a neuron’s identity from its population context with a permutation-equivariant transformer and transfers cell type and brain region to unseen animals [7]. It sees the activity itself. Our decoder sees only the correlation matrix, so it answers what the relations alone determine, and its target is stimulus tuning rather than cell class. This relational setting is common: recordings under natural behaviour have no controlled stimulus, and neurons from different sessions or animals share no stimulus-aligned coordinates, but they do share the structure of their correlations.

We answer this question with a decoder whose input is the correlation matrix of a sampled population, one row per neuron, and nothing else, and whose output is, per neuron, either a preferred orientation or a receptive-field position. It is trained on labelled populations and scored on neurons it never saw, at two scales. Within one brain we use the MICrONS release [8, 9]: 12,894 neurons from a cubic millimetre of mouse visual cortex, with in-vivo responses to a shared movie, a fitted digital-twin model [10], orientation labels for 5,287 of them and receptive-field labels for 11,326 (Fig. 1). The decoder trained on half the neurons reads the other half. Across brains we use the Allen Brain Observatory [11]: 33 mice recorded under identical stimuli. The decoder trained on some animals reads neurons of animals it never saw, which succeeds only when a sampled population mixes animals, and only with grating relations.

Averaging the correlations by orientation class shows that the relations between classes are almost circulant, which fixes the classes only up to a rotation or reflection of the orientation circle. A consistent departure from that symmetry fixes the absolute frame and makes the readout possible.

2 Methods

2.1 Data

MICrONS. The node table released with Ding et al. [8] holds 12,894 neurons co-registered to the electron-microscopy volume, across V1, RL, AL and LM, from 13 of the 14 two-photon scans: the authors’ working set after their quality restrictions, applied to the 13,952 matched neurons they report. Each neuron has a trial-averaged in-vivo response to the oracle natural-movie clips (120 bins), shown in every scan; a digital-twin response to a shared movie (4,999 bins, which we compress to 512 principal components; the Gram is preserved to $r > 0.999$); an in-vivo preferred orientation with a global orientation selectivity index (gOSI); and a receptive-field centre from the twin (spike-triggered-average fit, screen coordinates in $[-1, 1]$). The orientation label is the preferred orientation of the in-vivo tuning curve to Monet, a directional noise movie shown in 16 directions 22.5° apart in a separate block of the same scans, and gOSI is computed from that

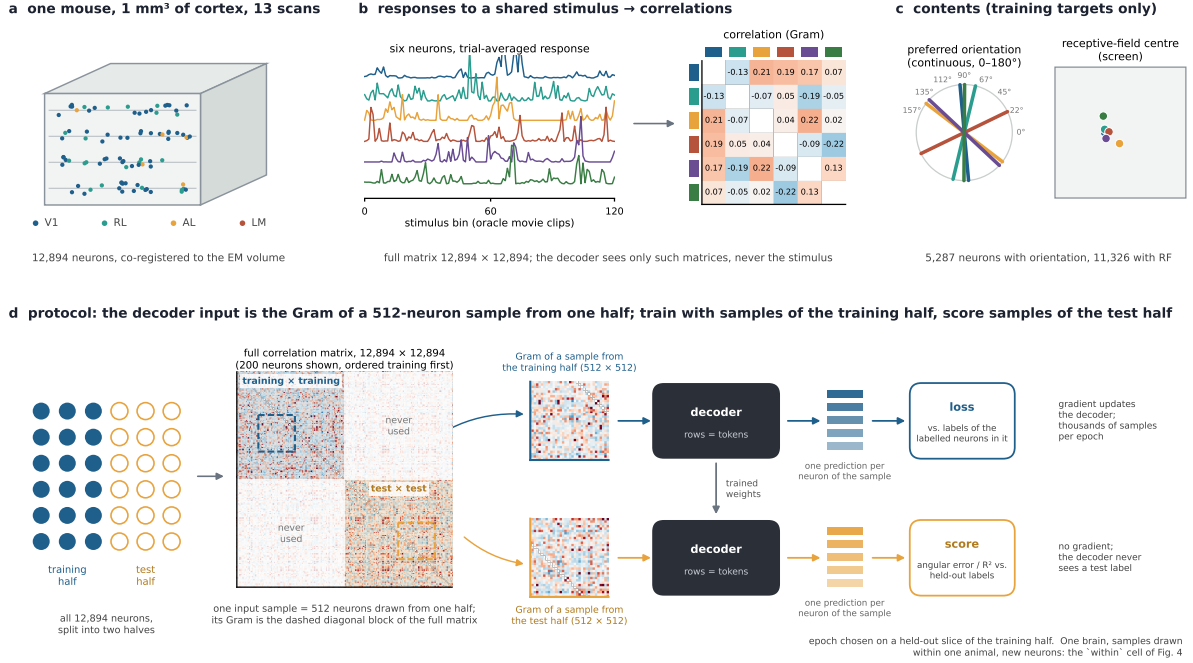


Figure 1: How the MICrONS task is built. (a) One mouse, 13 of the 14 two-photon scans of a cubic millimetre of visual cortex (areas V1, RL, AL, LM), 12,894 neurons co-registered to the electron-microscopy volume. (b) The relational substrate. Each neuron is its trial-averaged response vector to a stimulus all neurons saw (Methods); shown are six real neurons and their correlation matrix. The oracle clips were shown in all 13 scans, so the matrix is defined across scans. The decoder sees only such correlations, never the stimulus. (c) The contents of the same six neurons: preferred orientation (in vivo, a continuous angle, 5,287 neurons with $\text{gOSI} \geq 0.25$) and receptive-field centre (digital-twin fit, 11,326 neurons). (d) The protocol. Neurons are split into a training half and a test half; the full correlation matrix, ordered training-first (a real 200-neuron excerpt), is used only in its training×training and test×test blocks. A sample is 512 neurons drawn from one half; its 512 × 512 Gram (dashed) is the decoder’s entire input, one row per neuron, one prediction per row. Labels enter only the loss, never the input, and the halves share no neuron (Methods).

curve [8], so on MICrONS too the labels and the relations come from different stimuli. We use orientation, as a continuous angle, for the 5,287 neurons with $\text{gOSI} \geq 0.25$, and receptive-field position for the 11,326 neurons whose twin test correlation is ≥ 0.2 . The orientation labels are in-vivo measurements on both substrates (the release also carries a twin orientation, fitted the same way to the twin’s Monet responses, which we use only to bound label noise in Section 4 and never as a target); the receptive-field labels come from the twin’s fit on both substrates, so every receptive-field result, in vivo included, scores relations against model-derived positions (Section 4). Same-scan pairs correlate more strongly than cross-scan pairs at matched receptive-field distance: with the pairs of 6,000 receptive-field-labelled neurons binned by decile of centre distance, the same-scan mean is 1.4 to 1.6 times the cross-scan mean in vivo and 1.2 to 1.4 times on the twin over the six nearest deciles. The dependence of the correlation on receptive-field distance, and on the difference in preferred orientation (in vivo, 0.074 within and 0.061 across scans for pairs less than 15° apart), has the same sign and shape within and across scans; only the level differs. All populations mix scans, and the scan and area centring controls bound what scan identity contributes (Appendix B). Only 2% of the receptive-field label variance lies between scans and 9% between areas, so a decoder that read scan or area identity from the Gram could reach at most that much R^2 .

Allen. From Visual Coding 2P [11] we take the 36 VISp excitatory containers with at least 150 cells significantly responsive to drifting gratings: 33 mice (three contribute two containers each; splits are taken over mice, so all containers of a mouse fall on the same side of every held-out-animal split), 9,281 cells, of which 5,529 carry the static-grating orientation label used here; unlabelled cells stay in the populations and receive no loss, and the counts below are of all cells, labelled or not. The three held-out-animal splits have 6,706, 6,927 and 6,853 training cells against 2,575, 2,354 and 2,428 test cells; of the training cells, 1,746, 1,513 and 1,405 belong to the six further mice held out for epoch selection, leaving 4,960, 5,414 and 5,448 cells the decoder is trained on. Relations come from session A, either responses to natural movies shown to every mouse (4,500 bins) or the 40 blank-subtracted condition means of drifting gratings (8 directions $\times$ 5 temporal frequencies). Orientation labels come from static gratings in session B (six orientations 30° apart, $\text{gOSI} \geq 0.25$, $p < 0.05$), so labels and relations use different stimuli. Receptive-field centres come from locally sparse noise in session C (1,504 cells with a significant fit, 11 to 112 per mouse (median 37)).

2.2 Relations

Each neuron’s response vector is centred and normalised; the relation between two neurons is the cosine of their vectors, a signal correlation. For a population of n neurons the Gram is the $n \times n$ matrix of these values, standardised by an off-diagonal mean and standard deviation estimated once per run from the correlation matrix of 2,000 neurons drawn at random from the dataset, with the diagonal zeroed. These are two scalars, not per-neuron information; estimating them on the training neurons alone, or on each sampled population, moves them by at most 0.001 on either substrate. We call this standardised correlation matrix the Gram throughout.

2.3 Decoder

Each row of a population’s Gram, together with three row statistics (the mean, standard deviation and third moment of the row), is projected to a token; L pre-norm transformer blocks [12, 13] with H -head attention update the tokens, the Gram entering the attention logits as an additive bias; a linear head reads out, per token, the unit vector $(\cos 2\theta, \sin 2\theta)$ of the preferred orientation θ or an (x, y) receptive-field position. The loss is the mean squared error on the labelled tokens only; unlabelled neurons stay in the population and receive no loss. The standard configuration is width 512, 8 layers and 17M parameters with populations of 512 neurons on MICrONS, and

width 256, 4 layers and 2.2M parameters with populations of 256 neurons on Allen. Other sizes are in Appendix A. Training uses AdamW (weight decay 10^{-4}), a one-cycle schedule with peak learning rate 10^{-3} and 10% warm-up, on MICrONS 30 epochs of 5,000 populations in batches of 64 for orientation and 40 epochs for receptive fields, 4 attention heads and dropout 0.1; on Allen, 5,000 populations per epoch in batches of 64, for 30 epochs in the **within**, **pooledwithin** and **cross** regimes and 60 in **pooledcross**. Allen populations have 256 neurons except in the two single-animal regimes, which use 128, the largest size most animals can supply. Early stopping keeps the epoch with the lowest error on a selection slice: a random 20% of the labelled training half on MICrONS (528 of 2,643 neurons for orientation), whole mice held out of the training set on Allen. Width, depth, population size and the training budget vary between runs; the budget of every run is in its record, and the budgets behind Fig. A1 are given in its caption (Appendix A).

2.4 Protocol

Split. MICrONS neurons are halved at random. Training populations are sampled inside the training half and test populations inside the test half, so no test neuron appears in a training population (Fig. 1d). Early stopping uses a slice of the training half: the epoch with the lowest error on that slice is kept, so the test half never influences model selection.

Regimes (Allen). Two independent choices define a run. The split decides where test neurons come from: the training animals, each halved into training and test neurons, or held-out animals. The population decides what a sample is: neurons drawn from one animal, so that the Gram is a within-circuit matrix, or from several animals, so that most entries are between-animal correlations. We call the four regimes **within**, **pooledwithin**, **cross** and **pooledcross**. In every regime training populations are drawn from training neurons and test populations from test neurons, so no test population contains a neuron the decoder was trained on; in the training-animal regimes the test neurons come from animals some of whose other neurons were in training. A held-out-animal split holds out 9 of the 33 mice. Held-out regimes select the epoch on further held-out mice. Every held-out regime is run on three mouse splits and every training-animal regime on three neuron splits.

Metrics. Orientation is scored by the mean absolute angular error between the decoded angle, $\frac{1}{2} \text{atan2}$ of the output, and the label, taken modulo 180° , so it lies between 0° and 90° . A decoder whose output is independent of the label produces errors spread evenly over that range when either the labels or the outputs are uniform on the orientation circle, so chance is 45° . With the MICrONS label distribution the best constant prediction scores 41.3° (Appendix B). Allen labels sit on a 30° grid, so a decoder that recovered every true preference exactly would still disagree with them by 7.5° on average. A frame check reports the error after the best single rotation or reflection of all decoded angles, minimised over the two reflections and, for each, over a continuous offset. It is never above the raw error; we call the frame recovered when the raw error is within 5° of it and below 30° , and read a larger gap as structure without frame. Receptive fields are scored by $R^2 = 1 - \sum_i (\hat{y}_i - y_i)^2 / \sum_i (y_i - \bar{y})^2$, computed separately for the horizontal and the vertical coordinate and then averaged, with $\bar{y}$ the mean of that coordinate over the scored test cells and no scale or offset fitted. Labels are standardised by the mean and standard deviation of the training cells, so a decoder that predicts the training mean for every cell scores about 0 and one whose spread is at the wrong scale scores below 0. Every test neuron is predicted in many sampled populations and its predictions are averaged before the error is taken; the default averages about 37 randomly drawn populations per neuron, and wherever per-neuron predictions were saved exactly 32 covering partitions of the test half. The two differ by at most 0.3° , and numbers computed from saved predictions are marked as such in the tables.

Table 1: MICrONS, per-neuron decoding on the test half. Relations only: the 17M decoder on the Gram of 512-neuron populations, mean $\pm$ s.d. over 3 model seeds at neuron split 0. Balanced: the error averaged over eight equally weighted true-orientation bins. Per-neuron predictions were saved only for neuron splits 1 and 2, so the balanced column is that of the split-2 run, whose own raw error is 24.6° in vivo and 20.3° on the twin; within that run balancing costs $+3.3^\circ$ in vivo and $+4.2^\circ$ on the twin. The spread between neuron splits exceeds the spread between seeds: at model seed 0 the raw error is 26.0° , 25.4° , 24.6° in vivo and 20.6° , 19.3° , 20.1° on the twin over splits 0, 1 and 2. Stimulus-known linear readout: a linear map from each neuron’s own movie response vector to its content, a reference that uses the stimulus alignment the relational decoder discards, not a label-noise ceiling; the activity transformer of Table 4 is stronger on receptive fields (0.37 in vivo, 0.80 on the twin) (Appendix B). Chance: an uninformed decoder (45°) and the best constant orientation given the label distribution (41.3°), the reference for the raw error only; under the equal weighting of the eight bins every constant predictor scores 44.6° to 45.4° , so the reference for the balanced column is 45° .

content	substrate	relations only	balanced over orientations	stimulus-known linear readout	chance / best constant
orientation (angular error)	in vivo	$25.4^\circ \pm 0.5^\circ$	27.9°	20.2°	$45^\circ / 41.3^\circ$
orientation (angular error)	twin	$20.2^\circ \pm 0.4^\circ$	24.5°	15.8°	$45^\circ / 41.3^\circ$
receptive field (R^2)	in vivo	0.234 ± 0.007	–	0.34	0
receptive field (R^2)	twin	0.480 ± 0.010	–	0.68	0

2.5 Symmetry analysis

To expose the symmetry of the relations we average the Gram into a $K \times K$ class-Gram, the mean relation between neurons of two orientation classes, project it onto its circulant part, and ask which relabellings a held-out half’s class-Gram is compatible with (all $K!$ relabellings, Frobenius distance). A relabelling that differs from the truth by a rotation or a reflection of the orientation circle is an element of the dihedral group D_K . The analysis reports whether the truth is singled out absolutely or only modulo D_K .

3 Results

3.1 Per-neuron content from the correlation matrix alone

On MICrONS the decoder reads orientation and receptive-field position from the Gram of 512 neurons of the test half (Fig. 2, Table 1). The twin substrate is cleaner, since the twin removes trial noise. The predictions are in the true frame: the frame check lowers the error by less than 0.1° in every run. About half the neurons are decoded to within 15° (0.46 in vivo, 0.54 on the twin), but the error is not spread evenly over the orientation circle: the decoder resolves the cardinal axes and little else (Appendix B, Fig. 3). Asked only which cardinal axis a neuron is closer to, it is right for 87% of all neurons on the twin and 79% in vivo (majority answer 56%; averaged over the eight true-orientation bins, 83% and 74%), and for the neurons whose preference lies within 15° of an axis, for 94% on the twin and 86% in vivo, both values at each of neuron splits 1 and 2, the two splits whose predictions were saved. Those near-axis neurons are 1,445 and 1,482 of the 2,644 test neurons of the two splits. Sorted into four classes at 0° , 45° , 90° and 135° , 66% (twin) and 59% (in vivo) land in the right one against a majority class of 40%. Within 15° of a cardinal axis the error is 12° on the twin and 19° in vivo; within 15° of an oblique, 36° to 39° over both substrates, near chance (Fig. 3). Both readouts are properties of individual neurons, not of the scan: centring predictions and labels per scan leaves the receptive-field R^2 unchanged and the orientation error within 0.2° (Appendix B). Test Grams from stimulus bins the decoder never trained on add about 0.5° of error on the twin and 8.5° in vivo, where a Gram over 60 movie bins is partly specific to those bins (Appendix B).

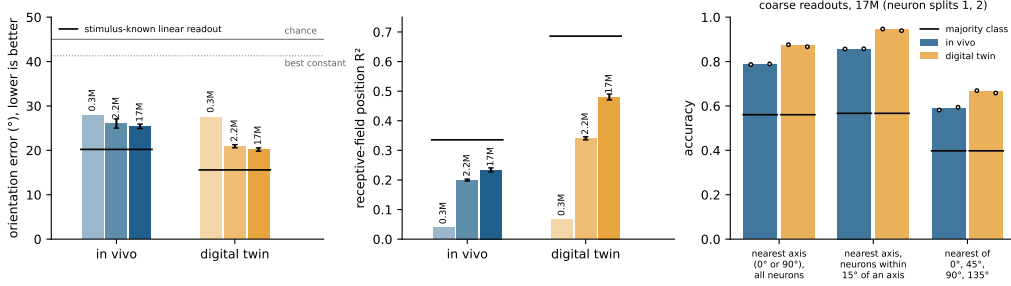


Figure 2: MICrONS: per-neuron content decoded from the population correlation matrix alone. Left: mean angular error of the decoded preferred orientation on held-out neurons of the same animal, for decoders of 0.3M, 2.2M and 17M parameters; error bars are the s.d. over 3 seeds; grey line: the 45° error of an uninformed decoder; dotted: the best constant, 41.3°; black marks: the stimulus-known linear readout of each neuron’s own responses. Middle: R^2 of the decoded receptive-field centre, same marks. Right: coarse readouts of the same decoder on neuron splits 1 and 2: whether the decoded angle falls on the correct cardinal axis, for all neurons and for the neurons whose preference lies within 15° of an axis, and whether it falls in the correct one of four classes at 0°, 45°, 90° and 135°; black ticks: majority class. The decoder sees only the standardised correlation matrix of 512 sampled neurons.

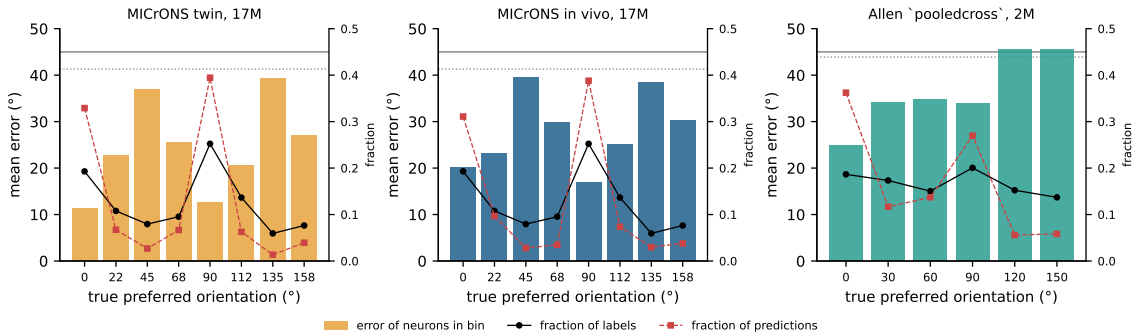


Figure 3: The readout is cardinal. Mean angular error of the neurons in each true-orientation bin (bars; grey line: 45° chance; dotted: the best constant under the raw error, not a per-bin reference, since a constant cardinal predictor scores about 6° in the bin it points at and about 85° in the orthogonal one), with the fraction of labels (black) and of predictions (red) per bin, for the twin and in-vivo MICrONS decoders (neuron split 2) and the Allen `pooledcross` decoder. Cardinal orientations are decoded well, obliques near chance, and most predictions fall on 0° or 90°.

3.2 Why the absolute frame is recoverable

The class-Gram of orientation is close to circulant (Fig. 4): neighbouring orientations correlate, orthogonal ones anti-correlate, and the pattern repeats around the circle. A circulant matrix is unchanged by every rotation and reflection of the orientation circle, so that part of the relations fixes the classes only up to the 16 elements of D_8 and cannot say which class is 0° . Whatever fixes the frame must be a departure from circulant structure, at the class level or finer. At the class level that residual holds 29% of the class-Gram variance in vivo and 19% on the twin over the whole volume (30% and 16% in V1), and it is signal: it correlates at 0.94 between random halves of the neurons on both substrates, at 0.84 (in vivo) and 0.99 (twin) between disjoint halves of the stimulus bins, and at 0.99 with the residual computed on class-balanced subsamples, so it is neither a finite-stimulus artefact nor a consequence of the uneven number of neurons per orientation. Its form is one-peaked, not bi-cardinal: in vivo, neurons preferring orientations within about 45° of 90° correlate more with each other than the circulant baseline predicts and the rest less, and on both substrates the same-class excess at 90° exceeds that at 0° (V1: +0.036 against -0.011 in vivo, +0.014 against +0.006 on the twin, where the gap between the two is smaller), and its shape differs between areas (residual correlation V1-RL 0.77, V1-AL 0.31).

Four observations narrow down what fixes the frame.

Priors do not explain the accuracy. The decoder reaches 20° to 25° , while predictors that use only the distribution of labels, the best constant orientation or each scan’s or area’s mean, reach 39° to 41° . Whether the frame comes from the label density is settled by the class-balanced populations below, which remove the prior from training and test alike and cost 1.8° .

The structure is recoverable with no frame in the labels. A decoder trained with each population’s targets rotated at random, and the loss taken after the best alignment of predictions to targets, places neurons to 19.9° on the twin and 25.5° in vivo after one global alignment, the same as the standard decoder with the frame (20.2° , 25.4°). This does not show that the circulant part alone carries the structure, since the residual is still in that decoder’s input. Its raw error, 25° and 31° against 45° for an arbitrary frame, shows that it fixes a consistent frame from the data even when the loss does not ask for one.

The frame is not the label density and not the particular sample of frames. With every training and test population drawn with equal numbers of neurons from each of the eight orientation classes, the decoder reaches 22.0° on the twin and 26.6° in vivo (two seeds each), with no gap between raw and aligned error. The frame also survives training and test relations built from disjoint halves of the stimulus, though both halves share the movie’s orientation content; on Allen, where the grating conditions cover orientation uniformly and the labels are nearly flat, the readout is cardinal all the same (Section 3.3), and it survives removing the three row statistics from the input (20.3°), while the statistics alone fix only a weak frame (37.7° ; Appendix B).

The errors are smallest at the cardinal axes and largest at the obliques. They are 11° to 12° near the cardinal axes and 37° to 39° in the two oblique classes on the twin, and reweighting the loss so that obliques count as much as cardinals moves them by at most 2.5° and does not close the gap (11° and 14° at the cardinals, 34° and 41° at the obliques, on neuron split 0 against split 2 for the standard run; Appendix B). The oblique ordering is not in the relations at single-neuron resolution.

In V1 the neurons of the dominant orientation correlate most strongly with each other. Define the anisotropy factor as the mean correlation between two neurons of the same orientation class, at the class where it peaks, divided by its mean over the other seven classes. In V1, where 28% of neurons fall in the 22.5° class centred on 90° , the factor peaks at 90° on both substrates (1.28 on the twin, 1.77 in vivo; Table 2), and 90° is the peak in 99% of neuron bootstraps on both substrates, and the values are unchanged on class-balanced subsamples, so the signature is in the correlations, not in the counts. The coincidence therefore holds in V1, which supplies 73% of the labelled neurons, and the whole volume, mostly the same neurons, peaks at 90° too, in 0.69 of bootstraps on the twin and 0.97 in vivo (Table 2); of the two smaller areas it holds in

AL on the twin, whose peak class is resolved in only 0.55 of neuron bootstraps, and in neither substrate in RL (0.39 and 0.73), so V1 is the only area with a bootstrap-resolved peak. The quantity is a signal correlation over trial-averaged responses, and on the twin over noiseless ones, so the anisotropy is one of tuning similarity, not of trial-to-trial co-fluctuation. Cardinal over-representation and its variation between areas are established features of mouse visual cortex [14, 15], and the preferential connectivity of like-tuned neurons [16] is one route by which their responses could become more similar.

Table 2: Label mode and anisotropy peak per area (MICrONS, 22.5° classes). Label mode: the class with the most neurons and its fraction. Peak: the class with the largest mean same-class correlation and the anisotropy factor (peak over the mean of the other seven classes). LM has 10 labelled neurons and is not given its own row; the volume row includes it. Bootstrap over neurons (2,000 resamples): 90% confidence interval of the factor at the peak class, and the fraction of resamples in which that class is the peak. V1 [1.21, 1.36], 0.99 (twin) and [1.63, 1.91], 0.99 (in vivo); RL [1.01, 1.45], 0.39 and [1.28, 2.15], 0.73; AL [1.25, 1.80], 0.55 and [1.12, 2.42], 0.38; whole volume [1.19, 1.33], 0.69 and [1.67, 1.95], 0.97.

area	neurons	label mode	peak, twin	peak, in vivo
V1	3,858	90° (0.28)	90° (1.28)	90° (1.77)
RL	1,125	0° (0.24)	112.5° (1.22)	112.5° (1.68)
AL	294	0° (0.32)	0° (1.49)	135° (1.69)
whole volume	5,287	90° (0.25)	90° (1.26)	90° (1.81)

A model population tests the candidate. We could not isolate what fixes the frame by subtraction, because removing the class-level residual from every Gram with the true classes injects the classes into the Gram (Appendix B), so the frame is located by exclusion and by a model (Appendix B, Table 6). Model neurons with von Mises tuning, a movie-like stimulus that covers orientation evenly, and the same pipeline give a class-Gram circulant to within noise, and the decoder trained on it recovers the ordering, 30° after one global alignment, but no frame, 54° raw. Sharper cardinal tuning and a cardinal excess of neurons fix the frame in at most one seed of five, alone, together, or at any strength tested; an excess of cardinal orientations among the stimulus frames does in three of five. Adding only the anisotropy the data show, a shared stimulus-driven response component among like-tuned neurons peaked at one orientation, with uniform counts and an even stimulus, fixes the frame in every seed at every strength run (15 of 15, 2° to 4° raw), from a residual weaker than the twin V1’s (0.05 of class-Gram variance against 0.16) to one between the twin’s and the in-vivo V1 value (0.23 against 0.30); every strength tested fixes the frame, so the runs give no lower bound on the strength required; the weakest setting is below cortex on the residual fraction and above it on the anisotropy factor (1.67 against 1.28 on the twin), so the model is neither uniformly weaker nor uniformly stronger than cortex. The same field with a uniform weight, so that no orientation is preferred, does not fix the frame (one seed, 41.6° raw against 40.4° aligned): its residual falls from 0.15 to 0.015 of the class-Gram variance and its peak class varies with the seed, so what fixes the frame is the peak of the weight, not the shared component as such. With training and test relations from disjoint halves of the frames, as on cortex, the peaked field still fixes the frame (two seeds, 7.0° and 16.6° raw) and the uniform one does not (49.0° raw, 40.0° aligned). The model residual has the one-peaked form of the cortical one and correlates 0.39 to 0.48 with it in vivo, against 0.94 for the cortical residual with itself across neuron halves. The model is far easier to decode than cortex (2° against 20° at a similar residual), so it shows that the anisotropy suffices, not that it accounts for the cortical error.

3.3 Orientation transfers between animals through mixed populations

With natural-movie relations, orientation is nearly absent from Allen correlations: same-orientation pairs correlate 0.005 more than orthogonal pairs, against 0.09 in MICrONS, and the

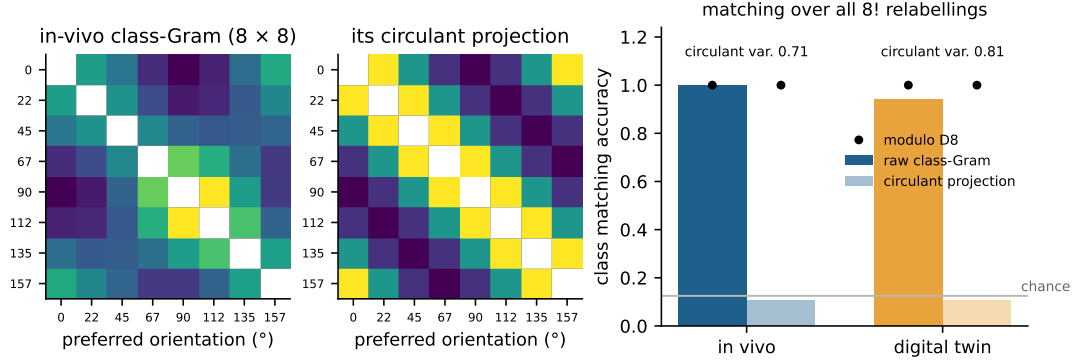


Figure 4: What fixes the absolute orientation frame. Left: mean correlation between orientation classes in vivo (diagonal masked). Middle: its circulant projection. Right: matching a held-out half’s class-Gram to the reference over all $8!$ relabellings. With the raw matrix the true labelling wins. Accuracy is the fraction of the eight classes given the correct label. After the circulant projection it falls to 0.11 on both substrates, against the chance level of $1/8$ (grey line), while accuracy modulo D_8 (dots) stays at 1.0, since a circulant matrix is invariant under D_8 : the matching then picks one of the 16 equivalent labellings, whose average per-class accuracy is $1/8$.

class-level tests are at or just above chance (Fig. A3). The Allen results below therefore use grating relations, a designed set of 40 conditions over an orientation grid, and do not replicate on naturalistic relations. With gratings the signal is present (0.05 within a mouse, 0.044 across mice) and the four regimes separate (Fig. 5, Table 3). Single-animal populations are close to uninformative: all six runs lie 1.9° to 2.9° below the 45° uninformed level, but five of the six are only 0.3° to 1.0° below the best constant of their own test set (42.5° to 43.9° per split) and the sixth is 0.4° above it. A margin of that size is inside the sampling noise of a test set of about 1,500 cells (bootstrap s.e. 0.9° over cells, 1.4° clustered on the nine test mice), so the six runs show a consistent direction, not a resolved effect, whether or not the animal was seen in training, and on every split the mixed populations of the same regime are 4.3° to 7.9° better (mixed populations of 128 neurons, matching the single-animal size, score 36.0° ; Appendix B). Mixed populations are 6.6° to 9.8° below chance in both rows: 36.2° on the training animals and 36.9° on held-out ones (the held-out mixed runs are trained for 60 epochs against 30 elsewhere; the training-animal row is budget-matched). The two rows use different splits, of neurons and of animals, so they cannot be differenced split by split; the three splits span 1.9° in the training-animal row (s.d. 1.0°) and 3.2° in the held-out row (s.d. 1.6°), against a difference of 0.7° between the row means, so three unpaired splits per row do not resolve a transfer cost of this size in either direction. As on MICrONS the readout is cardinal: over the three splits the decoder names the right cardinal axis for 62% of all cells and 68% of the cells labelled 0° or 90° (majority 51% and 52%; 61% averaged over the six label bins), cells labelled 0° are decoded to 28° and those labelled 90° to 36° , cells labelled 120° or 150° are decoded at or above the 45° chance level (49° and 51° pooled, 45° to 56° over the three splits), their predictions being pulled onto the cardinal axes, and the four-class readout is at the majority class (Fig. 3). What the mixed populations add is neurons from other animals, both as entries of the Gram and as diversity in the pool a population is drawn from; the controls below separate the two only partly. These exist only because all mice saw the same 40 conditions, and they measure how similar two tuning profiles are. The decoder still never sees which condition is which, so the absolute frame must again come from the population’s relational structure. Orientation is thus readable from a correlation geometry that spans animals, and only weakly from relations inside one circuit. Two controls locate the limit. The limit is not the neurons: with each neuron’s own 40-condition response vector as its token, the same decoder on the same single-animal populations reaches 33° to 35° , as with mixed populations (Table 3, last row). The limit is not the total number of labelled training cells either:

Table 3: Allen, mean angular error of the decoded orientation in the four regimes (2.2M decoder, 256-neuron populations; 128 for single-animal populations), three splits per regime (of neurons within each animal for the top row, of animals for the second row). The first three regimes are trained for 30 epochs and the held-out-animal mixed populations for 60, all at 5,000 populations per epoch in batches of 64, so the training-animal row is the budget-matched comparison; the budget of the stimulus-known reference row was not recorded. Last row: the same decoder on held-out animals with each neuron’s 40-condition response vector as its token instead of its Gram row, the stimulus-known reference. Chance 45° ; best constant of each run’s own test set 42.5° to 44.0° (43.7° over all 5,529 labelled cells); label grid 7.5° . Means over the three splits in parentheses.

test neurons from	single-animal populations	mixed populations
the training animals	$42.5^\circ / 42.9^\circ / 42.1^\circ$ (42.5°)	$35.3^\circ / 37.2^\circ / 36.1^\circ$ (36.2°)
held-out animals	$43.1^\circ / 43.1^\circ / 42.7^\circ$ (43.0°)	$35.2^\circ / 37.0^\circ / 38.4^\circ$ (36.9°)
held-out animals, stimulus-known reference	$32.8^\circ / 33.6^\circ / 34.9^\circ$ (33.8°)	$32.4^\circ / 33.7^\circ / 34.3^\circ$ (33.5°)

on held-out animals the single-animal and the mixed runs are trained on exactly the same cells (6,706, 6,927 and 6,853 by split), and on the training animals on 3,451 to 3,734 against 3,480. What differs is that each population is confined to one animal. An animal supplies 171 to 633 cells, 84 to 395 of them labelled. On held-out animals a training Gram is drawn from a whole animal; on the training animals each animal is halved, and a mouse whose training half holds fewer than the 128 neurons a population needs is skipped by the sampler, so the Grams come from the ten or eleven mice with pools of 128 to 333 cells, 1,904 to 2,160 cells in all. Either way the populations of one animal overlap in most of their neurons. Shrinking the pool instead, with mixed populations of 128 drawn from 280 or 560 training cells, costs 4.5° (40.5° and 40.7° against 36.0° from the full pool of 6,706; single runs, Appendix B), less than the 6° to 8° that confining populations to one animal costs. So roughly 4.5° of the gap is the size of the pool and 2° to 2.5° is specific to confining a population to one animal; the mixed populations drawn from a 280-cell pool still span animals, so no run here isolates the between-animal entries themselves. MICrONS separates the two: a decoder trained on 140 neurons of one scan and tested on the other scans scores 38° to 44° on the twin and 37° to 39° in vivo (three scans), while with the full training pool of one animal, single-scan populations of 128 are still decoded only to 32.0° on the twin and 31.1° in vivo, against 22.3° and 25.8° for mixed-scan populations of the same size (three seeds each; Appendix B). Confining populations to one scan therefore costs 5° to 10° even with the full training pool.

3.4 Receptive fields: what crosses animals is the animal’s screen position

Correlation falls with receptive-field distance within a mouse (0.127 at $< 5^\circ$ to 0.068 beyond 40°) and across mice (0.049 to 0.011). Decoding absolute screen coordinates on held-out mice gives R^2 that varies with the split, from 0.29 to -0.45 for single-animal populations and from 0.22 to -0.25 for mixed ones. Decomposing the saved predictions explains the variation (Fig. 6, Appendix C). The correlation between predicted and true mean position of each held-out mouse, pooled over 27 mouse-level predictions (9 mice per split, 20 distinct mice), is 0.36 to 0.71 ; shuffling the predicted means against the true ones inside each split gives $p \leq 0.04$, and restricting to the 20 distinct mice gives $r = 0.45$ to 0.80 at $p \leq 0.03$. The correlation between predicted and true position of a neuron relative to its mouse’s mean is -0.01 to 0.11 over 1,259 predictions of 898 distinct cells, each counted once per split in which its mouse is held out; the permutation shuffles the true relative positions inside each mouse and split. The decoder recovers the mean receptive-field position of an animal’s imaged patch, and the position of a neuron within that patch only weakly: the largest of the four within-mouse correlations, the vertical coordinate under mixed populations, is 0.11 with a permutation p of 0.001 against shuffles inside each mouse; the other three are 0.07 , 0.06 and -0.01 ($p = 0.02$, 0.07 and 0.6). R^2 on absolute coordinates

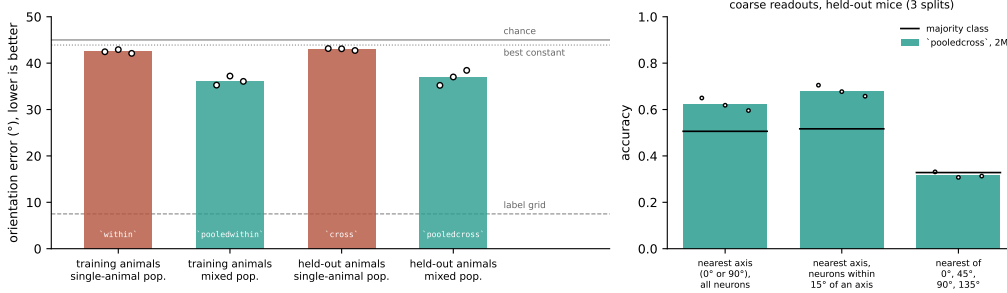


Figure 5: Allen, 33 mice, grating relations: orientation is readable only from populations that mix animals. Left: mean angular error; bars: mean over splits; points: the three splits; solid line: 45° chance; dotted: 43.9°, the best constant of the split-0 held-out test set and the largest of the twelve per-test-set values; each run’s own test set has a best constant of 42.5° to 44.0°, so the single-animal populations are not below their own baselines; dashed: the 7.5° label grid. Single-animal populations are 1.9° to 2.9° below chance whether the animal was seen in training or not; mixed populations are 6.6° to 9.8° below it in both rows. Right: coarse readouts of the mixed-population decoder on the held-out mice (three splits): correct cardinal axis for all cells and for the cells labelled 0° or 90°; the third bar, nearest of four classes 45° apart, maps each 30° label to one class (30° and 60° to 45°, 120° and 150° to 135°) and is shown for comparison with MICrONS; black ticks: majority class.

then depends mostly on whether the between-mouse spread of a particular test set is reproduced at the right scale.

The within-mouse layout is present in the correlations, since same-mouse correlation falls with distance. It is not learned. The most likely reason is that each mouse contributes only 11 to 112 receptive-field labels, but no control here varies the number of labels per mouse, and the within-mouse signal may simply be too weak: same-mouse correlation falls only from 0.127 to 0.068 over the whole distance range. With 11,326 labels in one animal, MICrONS lets the same architecture reach neuron-level receptive fields at $R^2 = 0.48$ with the 17M decoder, and 0.44 at the Allen decoder’s size of 2.2M (the 2.2M point of Fig. A1 reaches only 0.34 on the shorter schedule used for that curve).

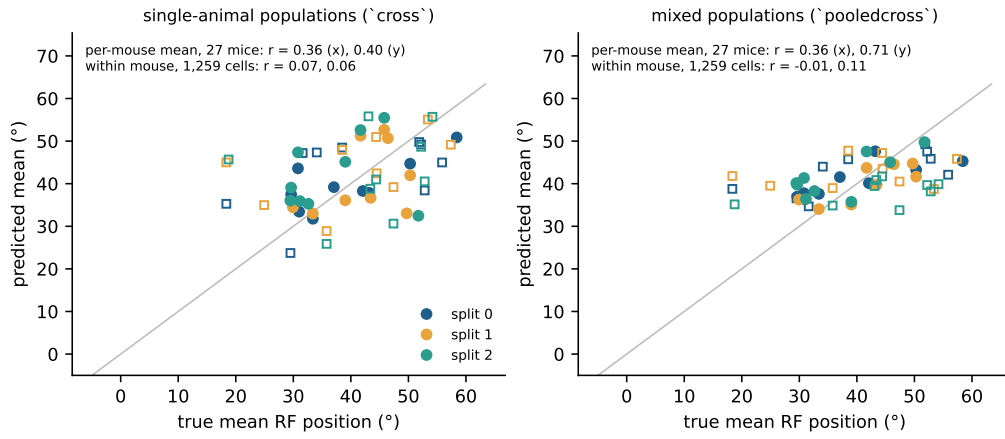


Figure 6: Allen receptive fields: what crosses animals is each animal's screen position. Predicted versus true mean receptive-field position of each held-out mouse (27 points over 3 splits (9 mice per split, 20 distinct mice); x squares, y dots), from decoders trained on single-animal (left) or mixed (right) populations. The per-mouse mean is recovered; the position of a neuron relative to its mouse's mean only weakly.

4 Discussion

A decoder trained on a disjoint set of labelled neurons, and given at test time nothing but the correlation matrix of a few hundred cortical neurons, predicts each neuron’s preferred orientation to a mean error of 25° in vivo and 20° on the digital twin, against 45° for an uninformed decoder and 41° for the best constant guess. The readout is cardinal: near-cardinal neurons to 12° to 19° , obliques near chance. It predicts each neuron’s receptive-field position to $R^2 = 0.23$ in vivo and 0.48 on the digital twin. The symmetry analysis constrains the mechanism: the relations fix the class structure up to the symmetries of the content, and a departure from that symmetry fixes the frame. That departure is a property of the correlations themselves: it is not the uneven number of neurons per orientation and not the particular stimulus sample. In V1 the neurons preferring the over-represented orientation have the most similar responses to one another, whether or not the sampled population is imbalanced. A direct ablation of that feature in the data is not available, but a model population with that anisotropy and nothing else recovers the absolute frame at every strength tested, while models with sharper or more numerous cardinal neurons do not; a cardinal excess among the stimulus frames also fixes it, in three seeds of five, and the orientation content of the movie was not measured.

Across animals the two contents behave differently. Orientation is carried by the relations between tuning profiles, which exist between any two neurons that saw the same conditions, in the same brain or not. From the correlations inside a single scan it is readable only weakly, at 31° to 32° on MICrONS, and inside a single Allen animal not measurably at all: each animal supplies 171 to 633 cells, so its populations of 128 neurons overlap in most of their neurons and differ too little from one another. Receptive-field position is carried at the level of an animal’s patch, and the neuron-level layout needs more labels per animal than these recordings provide.

Limits. MICrONS is one animal, and its training and test halves share recording sessions. The Allen held-out-animal cells are the cleaner generalisation test, but their variance across mouse splits is large, and they exist only with grating relations, a designed 40-condition set on which two tuning profiles can be compared, not with the naturalistic relations of the MICrONS result. The readout is cardinal: the obliques are not resolved on either dataset, so the mean error understates the precision at the cardinals and overstates it elsewhere. With training and test correlation matrices from disjoint halves of the movie the in-vivo error rises from 25.4° to 33.8° , against 25.3° for the same 60 bins on both sides, so about half the gain over the best constant is specific to the stimulus sample (Appendix B). No experiment removes the residual from the decoder’s input, so the assignment of the ordering to the circulant part and the frame to the residual is inferred from the symmetry argument and the model, not measured. The mechanism is established by exclusion and by a model, not by removing the candidate anisotropy from cortical data, since the direct ablation we tried injected the labels it was meant to remove (Appendix B). Orientation labels are noisy in both datasets (in vivo and twin labels agree on 70% of MICrONS neurons at 8 classes; static and drifting gratings on 45 to 67% of Allen cells), and the Allen labels are quantised to 30° , so 7.5° is a lower bound on every Allen error, relational and stimulus-known alike. The twin substrate is a model fitted to the same recordings, so the in-vivo results carry the claim about cortex and the twin results show what cleaner correlations would give; in AL the two substrates place the anisotropy peak two classes apart (0° on the twin, 135° in vivo), though the in-vivo peak is within noise of three other classes at 294 neurons. The receptive-field labels come from the twin for both substrates, so the twin receptive-field result compares a model’s correlations with the same model’s receptive fields, and the in-vivo result scores cortical correlations against model-estimated positions. The orientation labels are in-vivo measurements throughout.

The symmetry result is an instance of a general expectation: a representation that is equivariant to a group has a group-fixed part, identical across individuals up to relabelling, and an individual remainder [17]. The matching-over-relabellings test is the discrete form of unsupervised alignment of similarity structures across individuals by optimal transport [18]. This

paper adds the measurement in cortex, at the level of single neurons, and the finding that the absolute readout is possible and rests on a remainder that belongs to the correlations, not to the sample of neurons or stimuli.

Code and data. Code, run configurations and per-neuron predictions are in the directory `microns_ambiguity` (branch `microns-ambiguity`) of <https://github.com/entropicbloom/intentionality>. All data are public releases.

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A Scaling and population size

More capacity helps where labels are plentiful and not where they are scarce (Fig. A1). Twin receptive fields go from 0.34 to 0.48 to 0.51 R^2 between 2.2M, 17M and 57M parameters, but the budgets differ: the 2.2M point is 12 epochs of 2,500 populations and was still improving at its last epoch, the 17M and 57M points are 40 epochs of 5,000. The same 2.2M model trained on 80,000 populations reaches 0.44, and a 7.3M model on the same budget 0.45, so most of the 2.2M-to-17M gap is schedule, not capacity. Orientation error is flat from 2.2M upward on both MICrONS substrates (20.9°, 20.2°, 19.9° on the twin at 2.2M, 17M and 57M) and moves by 1.5° on Allen (35.2°, 36.5°, 36.7°; one run per point on one mouse split, against a 3.2° spread over the three mouse splits at 2.2M); only the 0.3M model is worse (27.5° on the twin and 27.8° in vivo). Every model above 2M memorises its training neurons within 10 to 20 epochs and is early-stopped on the held-out selection slice. Population size beyond 256 to 512 neurons changes orientation error by at most 2° in either direction, while twin receptive fields keep gaining slowly (Fig. A2). Split-to-split variance on held-out mice exceeds seed variance (Fig. A3).

B Controls

Controls on the MICrONS standard configuration (17M, 512-neuron populations, digital twin unless stated; Table 5).

What the decoder reads. With the three row statistics as the only input per neuron and no relational term, the error is 37.7°, raw and after alignment alike, so the scalars fix a weak frame on their own; with the Gram row alone and no statistics it is 20.3° on the twin and 26.5° in vivo, raw and aligned, so the relations carry the frame and the scalars are dispensable. (They are not a selectivity confound: their correlations with distance to the nearest cardinal axis are below 0.12 in magnitude and with gOSI below 0.16.) With the statistics plus the Gram entering only as an attention bias, which makes the decoder equivariant to the order of the neurons, the error is 19.9° over three seeds, the same as the standard decoder (20.2°), whose input layer also projects the raw Gram row in sample order.

Reference decoders that see the activity. To measure what the relations discard, the same transformer was trained with each neuron’s token built from its response vector instead of its Gram row, so that it sees the stimulus-aligned activity the Gram is computed from (Table 4). Adding the Gram to it as an attention bias changes nothing. A linear readout of each neuron’s own response vector, with no population context, does as well as the activity transformer on orientation (20.2° against 21.5° in vivo, 15.8° against 15.0° on the twin), so on orientation the activity decoder is a per-neuron tuning readout whose only advantage over the relational decoder is the alignment of the responses to the stimulus. On receptive fields the transformer is ahead of the linear readout (0.37 against 0.34 in vivo, 0.80 against 0.68 on the twin), so there the population context adds something. Against the activity transformer, the relations retain about 80% of the gain over the best constant on orientation and about 60% of the R^2 on receptive fields (against the linear readout, about 80% and about 70%); the twin receptive-field gap is inflated by the twin’s labels and responses coming from the same model. With test responses from movie frames the decoder never trained on, the activity decoder has no valid input, since its coordinates now index different frames, and it falls to the level of the best constant or below it (49.5° in vivo, 41.0° on the twin, against 41.3° for the best constant), while the relational decoder scores 33.8° in vivo and 20.7° on the twin. This shows which inputs each readout can be applied to, not which readout is better. The activity decoder transfers across animals under a shared stimulus protocol (Section 3.3) but not across stimuli. The relations do both, because a correlation over enough frames depends only weakly on which frames it was computed from, and mainly on both

neurons having seen the same ones (0.5° over 2,500 bins, 8.5° over 60; Cross-stimulus test). A decoder given the response vectors with the stimulus bins shuffled, the same shuffle for every neuron of a population, is blind to the stimulus but free to compute any relation among the neurons. It is no better than the Gram decoder on orientation and far below it on receptive fields (Table 4), but it has to discover the relations under a fresh permutation for every population, so the comparison does not show whether richer statistics than the pairwise Gram exist.

Table 4: Reference decoders that see the activity (each neuron’s stimulus-aligned response vector), 17M unless linear, 512-neuron populations. Orientation as mean angular error (chance 45°), receptive field as R^2 ; mean $\pm$ s.d. over three seeds where given, otherwise seed 0.

decoder input	ori. in vivo	ori. twin	RF in vivo	RF twin
Gram row (this paper)	$25.4^\circ \pm 0.5^\circ$	$20.2^\circ \pm 0.4^\circ$	0.23 ± 0.01	0.48 ± 0.01
activity, transformer	$21.5^\circ \pm 0.4^\circ$	$15.0^\circ \pm 0.1^\circ$	0.37	0.80
activity + Gram as attention bias	22.2°	15.0°	–	–
activity, linear, no population context	$20.2^\circ \pm 0.2^\circ$	$15.8^\circ \pm 0.1^\circ$	0.34 ± 0.01	0.68 ± 0.01
activity, transformer, disjoint stimulus bins	49.5°	41.0°	–	–
activity, bins shuffled per population, transformer	$26.3^\circ \pm 0.6^\circ$	$27.8^\circ \pm 6.5^\circ$	0.04	0.08
activity, bins shuffled, linear, no population context	40.3°	40.2°	–	–

Cross-stimulus test. Training and selection Grams built from one random half of the stimulus bins and test Grams from the other half give 20.7° on the twin (2,500 bins per half, raw responses rather than principal components; three seeds, 20.3 to 21.2) and 33.8° in vivo (60 bins per half; 33.4 to 34.2), against 20.2° and 25.4° with shared bins. A control with the same 60-bin half on both sides gives 25.3° , so the in-vivo increase comes from the stimulus sample, not the smaller Gram: a correlation matrix over 60 movie bins partly reflects which frames it was built from, and one over 2,500 bins does not.

Scan and area. The MICrONS neurons come from 13 scans in four areas, same-scan pairs correlate more strongly than cross-scan pairs, and different scans cover different parts of the screen, so a decoder that read scan identity from the Gram could predict each scan’s mean receptive-field position and score a positive R^2 without knowing any neuron’s position. The Allen decoder does exactly that (Section 3.4), so we checked MICrONS the same way, centring predictions and labels per scan and per area before scoring. The receptive-field result holds: on a further run of the same architecture at seed 0 on the shorter 30-epoch schedule, the only receptive-field run with saved predictions, R^2 recomputed from those predictions is 0.499 absolute and 0.497 within scan on the twin, 0.246 and 0.253 in vivo, and 0.47 (twin) and 0.22 (in vivo) within area. Only 2% of the label variance lies between scans and 9% between areas, so a scan-identity readout could reach at most 0.02. For orientation, a predictor that assigns every neuron its scan’s or area’s mean orientation scores 39° , and removing a separate rotation per scan or area changes the decoder’s error by less than 0.2° . Decoders trained on V1 alone transfer to the other areas unevenly across seeds (Table 5): to AL the aligned error is 32° in all three seeds but only one seed lands on a rotated frame, and to RL one seed transfers and two do not. The RL row uses 512-neuron populations and the V1 and AL rows 128. With class-balanced training populations and unrestricted test populations, V1 itself is decoded 9° worse than with the standard training set (31° against 22.4° , two seeds against three, populations of 128). This is not the same control as the whole-volume one, where training and test populations are both balanced and the cost is 1.8° : the V1 run leaves a cardinal-heavy test set in place and is trained on a smaller pool. V1’s test set is the more cardinal-heavy of the two (29% at 90° against 26%, best constant 37.1° against 40.3°), which accounts for about 3° of the difference; the whole-volume run is the one that bears on the claim of Section 3.2. Retinotopy is a finer prior than scan or area, and the Gram carries receptive-field position, so we bounded it directly. Over the 4,835

neurons with both labels, a cross-validated k -nearest-neighbour predictor of orientation from the twin receptive-field centre reaches 39.1° to 41.2° ($k = 20$ to 200), no better than the 39° scan-mean predictor and far above the decoder’s 20° to 25° . The anisotropy is not a retinotopic artefact either: with same-class pairs reweighted to the same distribution of receptive-field-centre distances as the other classes, V1’s factor at 90° is 1.33 on the twin and 1.80 in vivo, against 1.28 and 1.77 unmatched, and 90° remains the peak in every distance band.

Class balance and priors. Preferred orientations are uneven in MICrONS (19% of labelled neurons within 11° of 0° , 25% of 90° , 6 to 10% in each oblique bin), so a decoder could gain from the distribution alone. It gains little: the best constant prediction has an error of 41.3° , a predictor that assigns every neuron the circular mean of its scan 39.0° and of its area 38.7° , against 20.3° for the decoder on the twin and 24.6° in vivo (neuron split 2). Training and testing on populations drawn with equal numbers of neurons from each of the eight classes (64 per class) gives 22.0° on the twin and 26.6° in vivo (two seeds: 22.4, 21.5; 26.5, 26.7), raw and after global alignment alike, so the absolute frame does not depend on the class imbalance. The per-area errors are within 3° of each other: the frame is read from a volume-wide sample, so an area whose label mode differs from the volume’s peak is not mis-framed. On Allen the labels are nearly flat (14 to 20% per orientation), the best constant is 43.9° , and the balanced error is 36.4° against 35.7° raw on the saved per-neuron predictions of held-out-animal split 0, which average 32 population draws per neuron (Table 3 reports the population-level error of the same run, 35.2°). The per-bin errors show, on both datasets, that the readout is cardinal: neurons preferring 0° or 90° are decoded to 11° to 12° on the twin and, on held-out-animal split 0, to 25° and 34° on Allen; oblique ones to 37° to 39° on the twin and, on the same split, to 34° and 35° at 30° and 60° but 45° at 120° and 150° (pooled over the three splits the Allen values are 28° , 36° , 31° , 34° , 49° and 51° ; Section 3.3), and the predictions pile up on the two cardinal axes: 63 to 73% of MICrONS predictions fall within 11.25° of 0° or 90° (two neuron splits per substrate) and 51 to 63% of Allen predictions within 15° (three mouse splits), the half-widths of the bins of Fig. 3. The relations say how close a neuron is to a cardinal axis, which is what the anisotropy of Section 3.2 predicts, and much less about which oblique it prefers.

A subtraction ablation that leaks labels. To remove the class-level anisotropy from the input, we estimated the class-Gram residual (8 or 24 classes) on the labelled training half and subtracted the class-pair offset from every Gram entry, using each neuron’s true class, at training and test. The twin error was unchanged (20.7° raw and aligned at 8 classes) and improved at 24 classes (17.7°); the control that subtracts the whole class structure, which should destroy decoding, improved it further (17.2° raw, 15.6° aligned). An offset keyed to the true class encodes that class in the Gram, through its jumps at class boundaries and its estimation noise, and the decoder reads it. No result of this ablation is used.

Synthetic populations. 6,000 model neurons with von Mises tuning ($\kappa = 2$), preferred orientations either uniform or with half the neurons within about 12° of a cardinal axis (count), tuning either isotropic or sharpened to $\kappa = 4$ at the cardinals (sharp), and a 120-frame stimulus that either covers orientation evenly or has half its frames near a cardinal axis (stim); responses with Gaussian noise; the 2.2M decoder on 512-neuron populations, epoch selected on the raw error as in every other run, five data and decoder seeds (Table 6). With every switch off the class-Gram is circulant to within noise (neighbour-correlation range 0.01 against 0.04 to 0.14 with a switch on), and the decoder recovers the ordering but no frame. Sharper cardinal tuning and a cardinal excess of neurons fix the frame in at most one seed of five, alone, together, or at any strength (κ up to 32; gain up to 5 times; two seeds each, all above 42° raw); stimulus anisotropy fixes it in three of five seeds, and in four of five with sharper cardinal tuning. The anisotropy the cortical data show (Section 3.2) is modelled as a shared stimulus-driven component among

Table 5: Controls, mean angular error on held-out neurons (chance 45°), 17M decoder on 512-neuron populations unless stated; single runs unless a seed count is given. Where two numbers are given, the second is the error after the best global rotation or reflection (rows marked *aligned*) or averaged over eight equally weighted true-orientation bins (rows marked *balanced*).

control	substrate	error
standard decoder, split 0 (3 seeds)	twin	20.2°
row statistics only, no relational term	twin	37.7°
Gram row only, no row statistics	twin / in vivo	$20.3^\circ / 26.5^\circ$
row statistics + Gram as attention bias (3 seeds)	twin	$19.9^\circ \pm 0.3^\circ$
disjoint stimulus bins, training vs test (3 seeds)	twin	$20.7^\circ \pm 0.4^\circ$
standard decoder, split 0 (3 seeds)	in vivo	25.4°
disjoint stimulus bins (60 per side; 3 seeds)	in vivo	$33.8^\circ \pm 0.3^\circ$
same 60 bins on both sides	in vivo	25.3°
neuron split 1 / 2, single runs	in vivo	$25.4^\circ / 24.6^\circ$
rotated targets, frame-free loss, raw / aligned	twin	$25.4^\circ / 19.9^\circ$
rotated targets, frame-free loss, raw / aligned	in vivo	$30.9^\circ / 25.5^\circ$
standard decoder, split 2, raw / balanced	twin	$20.3^\circ / 24.5^\circ$
orientation-balanced loss, split 0, raw / balanced	twin	$21.4^\circ / 25.3^\circ$
standard decoder, split 2, raw / balanced	in vivo	$24.6^\circ / 27.9^\circ$
orientation-balanced loss, split 0, raw / balanced	in vivo	$26.0^\circ / 29.1^\circ$
Gram row only, saved predictions, raw / balanced	twin / in vivo	$20.5^\circ / 25.2^\circ, 26.2^\circ / 29.6^\circ$
class-balanced populations (64 per class, 2 seeds), raw / aligned	twin	$22.0^\circ / 21.9^\circ$
class-balanced populations, raw / aligned, 2 seeds	in vivo	$26.6^\circ / 26.4^\circ$
trained on V1, scored on V1 (128), 3 seeds	twin	$22.4^\circ \pm 0.5^\circ$
trained on V1, scored on RL (512), raw / aligned, 3 seeds	twin	26, 43, 46 / 25, 41, 39
trained on V1, scored on AL (128), raw / aligned, 3 seeds	twin	58, 33, 32 / 32, 32, 32
same, class-balanced training (2 seeds), V1 / RL / AL, raw	twin	$31^\circ / 29^\circ / 45^\circ$
aligned	twin	$31^\circ / 29^\circ / 36^\circ$
within-scan populations of 128, 3 seeds	twin / in vivo	$32.0 \pm 1.1^\circ / 31.1 \pm 1.2^\circ$
one training scan (140 neurons), populations of 128	twin / in vivo	$38^\circ\text{--}44^\circ / 37^\circ\text{--}39^\circ$
mixed-scan populations of 128, 3 seeds	twin / in vivo	$22.3 \pm 0.4^\circ / 25.8 \pm 0.2^\circ$
Allen, mixed populations of 128, pool 280 / 560 / 6,706 cells	2.2M	$40.5^\circ / 40.7^\circ / 36.0^\circ$

like-tuned neurons: a random component per frame on 36 orientation nodes, smoothed over the circle with a von Mises kernel ($\kappa = 4$, half-width about 30°) so that only neurons with nearby preferences share it, enters each neuron’s response with a weight peaked at 90° , with uniform counts, isotropic tuning and an even stimulus. Its class-Gram residual has the one-peaked form of the cortical one and correlates 0.39 to 0.48 with the in-vivo V1 residual; its size is set by the weight and reported as the residual’s fraction of class-Gram variance (V1: 0.16 on the twin, 0.30 in vivo) together with the anisotropy factor (1.67, 1.99 and 1.92 at the three sizes). A global field over orientation with three Fourier modes was run at four strengths: it fixes the frame in two of five seeds at anisotropy factor 1.09 and residual fraction 0.00, four of five at 1.32 and 0.01, and five of five at 1.68 and 0.06 and at 1.75 and 0.07; its residual correlates negatively with the cortical one (-0.38), so the local field is the model reported. The spread across seeds is large in every row where the frame is not fixed, because selection on the raw error then has no signal to select on, and the row means are therefore reported with their range. The frame-fixed counts are unchanged with the raw-error threshold at 25° or 35° , except that the two rows with a cardinal excess of neurons and no stimulus anisotropy, count and count + sharper tuning, drop to 0 of 5 at 25° (their single frame-fixed seeds are at 27.1° and 29.4° raw). One run (all three switches, seed 0) never left its initial loss and recovered neither ordering nor frame; it is included in the mean.

Table 6: Synthetic populations, five seeds: raw error / error after the best global rotation or reflection, mean over seeds with the range in brackets, and the number of seeds in which the frame was fixed (raw error below 30° and within 5° of the aligned error). Bottom block: a shared stimulus-driven component among like-tuned neurons (a local random field over preferred orientation, weight peaked at 90°), at three sizes of the class-Gram residual it produces (V1: 0.16 twin, 0.30 in vivo), with training and test relations from disjoint halves of the frames, and with a uniform weight, which gives a residual of 0.015 and no fixed peak class. A gap between raw and aligned error means structure without frame.

anisotropy switched on	raw	aligned	frame fixed
none (circulant)	53.7° [44–63]	30.0° [23–42]	0 / 5
stimulus	23.5° [4–54]	13.9° [4–19]	3 / 5
sharper cardinal tuning	51.4° [39–65]	28.6° [15–42]	0 / 5
cardinal excess of neurons (count)	37.8° [27–45]	35.1° [27–42]	1 / 5
sharper tuning + stimulus	12.2° [3–38]	6.3° [3–10]	4 / 5
count + stimulus	25.0° [4–47]	20.7° [4–43]	3 / 5
count + sharper tuning	42.8° [29–58]	35.2° [27–41]	1 / 5
all three	18.4° [3–46]	17.9° [3–44]	4 / 5
shared component among like-tuned neurons, residual 0.05	3.3° [3–4]	3.2° [3–4]	5 / 5
residual 0.15 (twin V1: 0.16)	2.4° [2–3]	2.4° [2–3]	5 / 5
residual 0.23 (in vivo V1: 0.30)	2.1° [2–2]	2.0° [2–2]	5 / 5
residual 0.15, disjoint frame halves (2 seeds)	11.8° [7–17]	11.6° [7–16]	2 / 2
uniform weight, residual 0.015 (1 seed)	41.6°	40.4°	0 / 1
uniform weight, disjoint frame halves (1 seed)	49.0°	40.0°	0 / 1

C Receptive-field decomposition per split

Per held-out-mouse split, for the two Allen regimes of Fig. 6: R^2 on absolute screen coordinates, the correlation between predicted and true per-mouse mean position over the 9 test mice (x, y), and the correlation between predicted and true position relative to the mouse mean over the cells (x, y). The values quoted in Section 3.4 pool the three splits: 27 mouse-level predictions per regime and 1,259 cell-level predictions (898 distinct cells of the 1,504 with a significant fit, each counted once per split in which its mouse is held out).

Table 7: Allen receptive fields per split.

populations	split	absolute R^2	per-mouse mean $r(x, y)$	within-mouse $r(x, y)$
single-animal	0	0.29	0.47, 0.74	0.07, 0.11
single-animal	1	-0.02	0.48, 0.40	0.06, 0.08
single-animal	2	-0.45	0.19, 0.25	0.11, -0.01
mixed	0	0.22	0.70, 0.81	0.00, 0.20
mixed	1	0.08	0.23, 0.80	0.02, 0.08
mixed	2	-0.25	0.46, 0.75	-0.06, 0.04

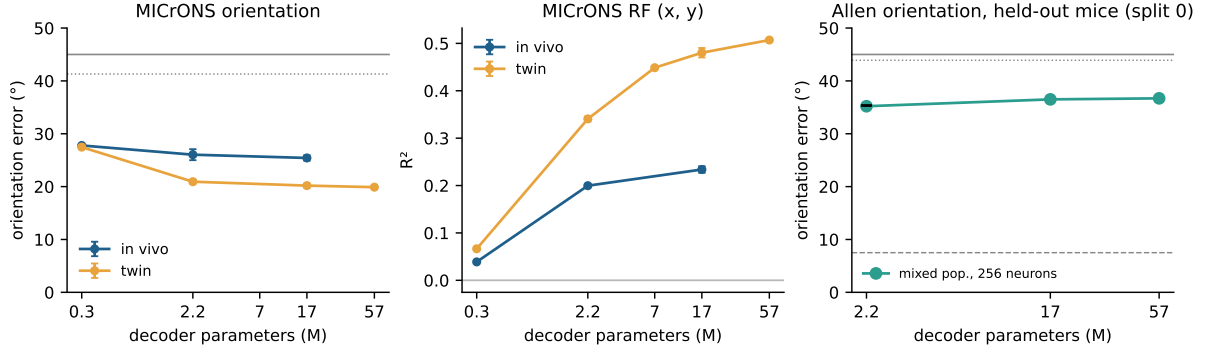


Figure A1: Scaling with decoder size. Test metric versus number of decoder parameters at fixed population size (512 neurons MICrONS, 256 Allen). Error bars: s.d. over 3 seeds where run. On MICrONS receptive fields R^2 rises to 57M, but only the 17M and 57M points share a training budget; the orientation error is flat from 2.2M upward on both MICrONS substrates and moves by 1.5° on Allen, one run per point on one mouse split. The receptive-field points do not share a training budget: 20,000 populations at 0.3M, 30,000 at 2.2M, 80,000 at 7.3M and 200,000 at 17M and 57M.

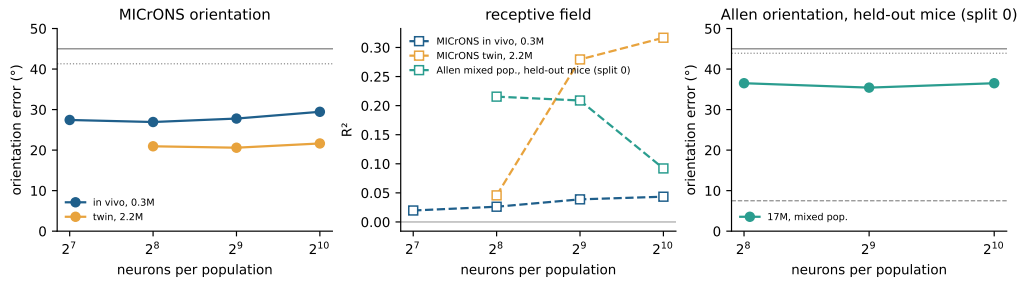


Figure A2: Population size. Test metric versus the number of neurons in each sampled population. Beyond 256 to 512 neurons the orientation error moves by at most 2° , up or down; twin receptive fields keep gaining slowly, Allen receptive fields lose.

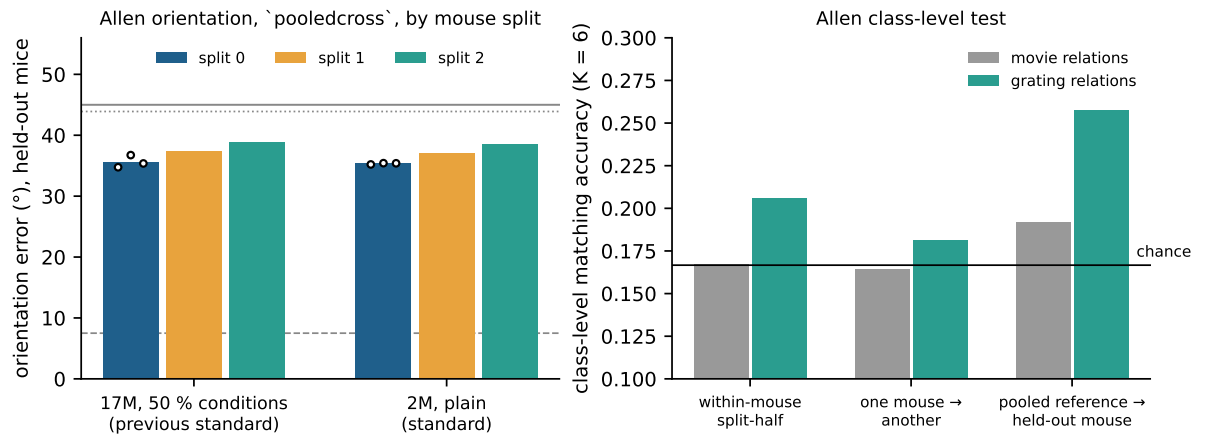


Figure A3: Allen split variance and the class-level test. Left: pooledcross orientation error on the same three held-out-mouse splits for the previous standard configuration (17M, with a training-time Gram augmentation no longer used) and the final one (2.2M); split 0 carries three seeds. Right: the class-level test with relations from natural movies versus drifting gratings; only the grating relations move every test off chance.